Unit DoF10

SKU:U230

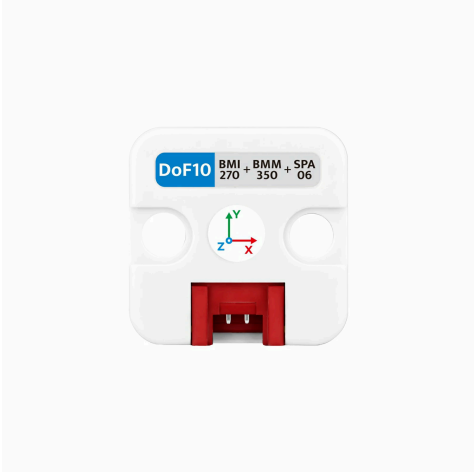






















Description

Unit DoF10 is a 10-axis attitude sensing unit that integrates a six-axis inertial sensor, a three-axis magnetometer, and a barometric pressure sensor. It features the low-power BMI270 IMU, the BMM350 three-axis magnetometer, and the SPA06-003 pressure sensor. The BMI270 integrates a 16-bit 3-axis accelerometer and a 16-bit 3-axis gyroscope, with configurable measurement ranges and on-chip motion event detection. The BMM350 uses TMR (tunnel magnetoresistance) technology for three-axis magnetic field measurement, achieving approximately 0.1 µT resolution and excellent temperature stability. The SPA06-003 measures pressure, temperature, and relative altitude.

Unit DoF10 outputs acceleration, angular velocity, temperature, three-axis magnetic field, and pressure data in real time, making it suitable for robot attitude estimation, heading detection, electronic compass, motion tracking, barometric altimetry, floor change detection, drones, and other applications.

Features

- Integrated six-axis inertial, three-axis magnetic, and barometric pressure sensors
- I2C communication interface
- BMI270 with 16-bit sampling and motion detection
- BMM350 with TMR technology and high magnetic field resolution
- SPA06-003 for pressure, temperature, and altitude measurement
- Supports various attitude, heading, and altitude applications

Includes

- 1 x Unit DoF10
- 1 x HY2.0-4P Grove Cable (5cm)

Applications

- Robot attitude sensing
- Heading detection
- Electronic compass
- Motion tracking
- Barometric altimetry
- Floor change detection
- Drones

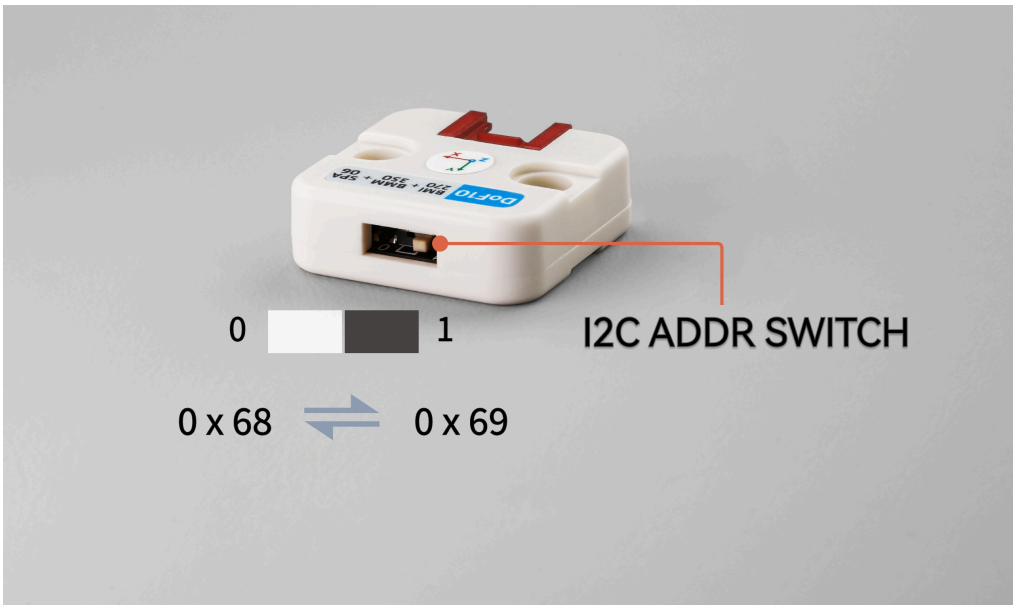
Specifications

Specification	Parameter
IC Model	BMI270 + BMM350 + SPA06-003
Sensor Type	Six-axis inertial, three-axis magnetic, and barometric pressure sensor
Accelerometer	16-bit 3-axis accelerometer
Accelerometer Range	±2 g / ±4 g / ±8 g / ±16 g
Accelerometer Accuracy	±20 mg (approx. ±0.02 g)
Gyroscope	16-bit 3-axis gyroscope
Gyroscope Range	±125 / ±250 / ±500 / ±1000 / ±2000 °/s
Gyroscope Accuracy	±0.5 °/s
Magnetometer	BMM350 three-axis magnetometer
Magnetic Field Range	X/Y/Z axes: ±2000 μT (typical on each axis)
Magnetic Field Resolution	approx. 0.1 μT
Barometric Pressure Sensor	SPA06-003
Pressure Measurement Range	300 ~ 1100 hPa
Absolute Pressure Accuracy	±0.3 hPa (typical)
Relative Pressure Accuracy	±0.03 hPa (typical)
Temperature Measurement Range	-40 ~ +85 °C
Temperature Accuracy	±1 °C (typical)
Communication Interface	I2C
BMI270 I2C Address	I2C @ 0x68 / 0x69 (switchable via onboard switch)
BMM350 I2C Address	I2C @ 0x14
SPA06-003 I2C Address	I2C @ 0x76
Product Size	24.0 x 24.0 x 8.0mm
Product Weight	3.1g
Package Size	138.0 x 93.0 x 7.0mm
Gross Weight	6.6g

Learn

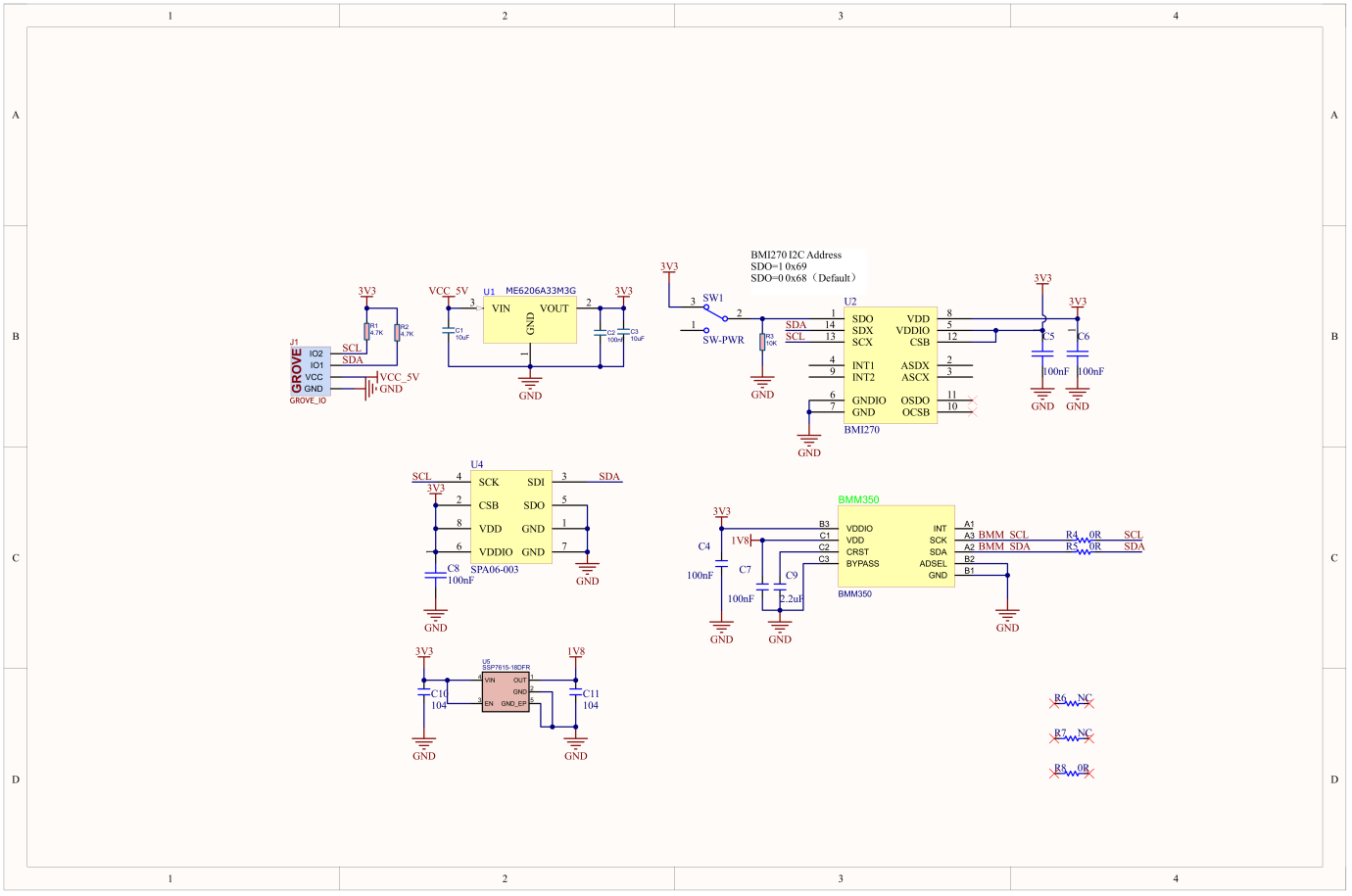
I2C Address Switching

The BMI270 I2C address can be switched with the onboard switch. Set the switch to `0` for address `0x68`, or to `1` for address `0x69`, to avoid address conflicts on the I2C bus.



Schematics

- Unit DoF10 Schematics PDF



PinMap

Unit DoF10

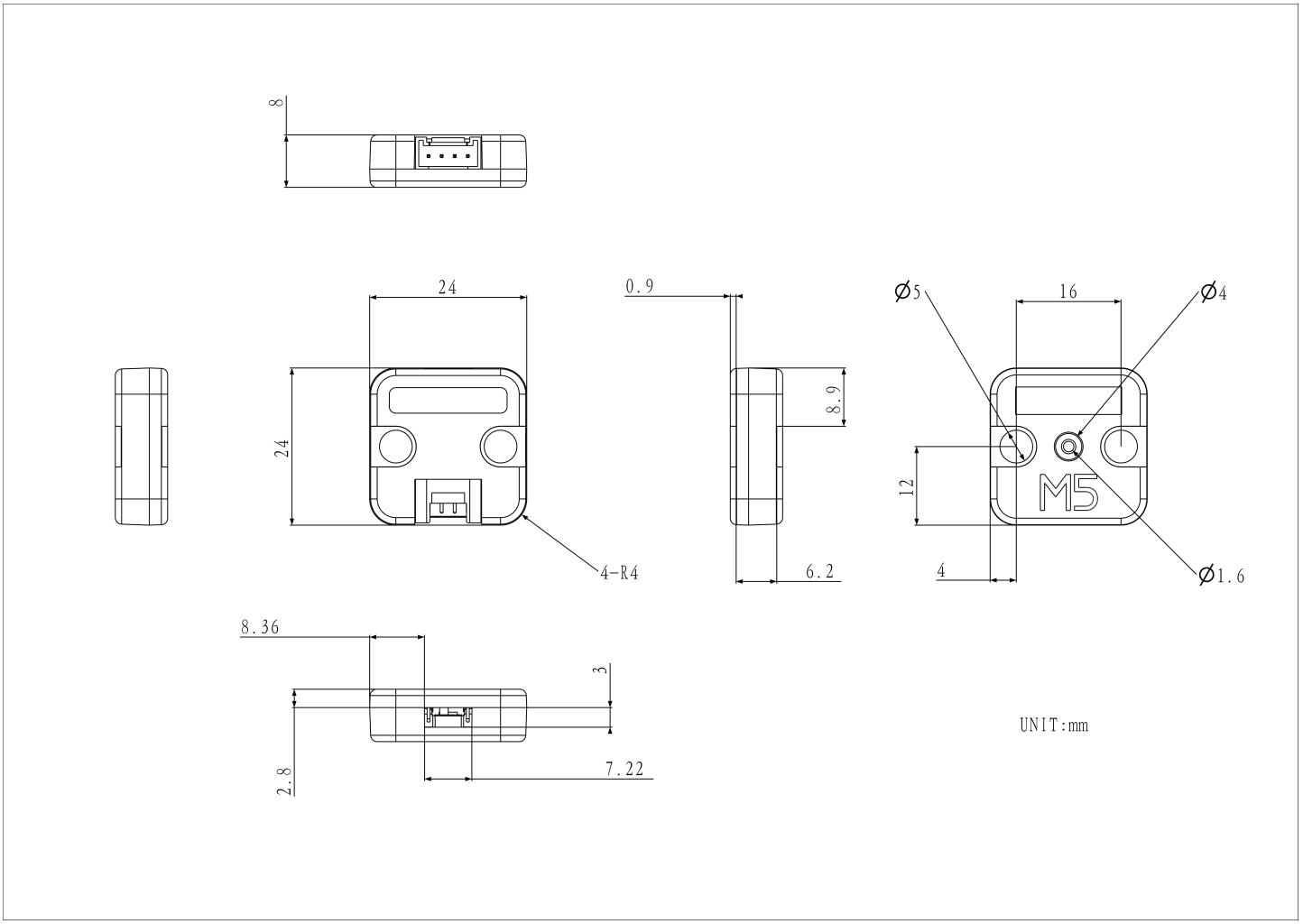
HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Datasheets

- BMI270 Datasheet
- BMM350 Datasheet
- SPA06-003 Datasheet

Model Size

- Unit DoF10 Model Size PDF



Structure

- Unit DoF10 Structure Files

Softwares

Arduino

- Unit DoF10 Arduino Quick Start
- Unit DoF10 Arduino Library
- M5Unified
- M5UnitUnified

UiFlow2

- Unit DoF10 UiFlow2 Docs

Video

Product Comparison

Product Comparison				
	Unit DoF6	**Unit DoF9**	**Unit DoF10**	**Unit BMM350**
Sensor Combination	BMI270	BMI270 + BMM350	BMI270 + BMM350 + SPA06-003	BMM350
Sensor Type	Six-axis inertial sensor	Six-axis inertial + three-axis geomagnetic sensor	Six-axis inertial + three-axis geomagnetic + barometric pressure sensor	Three-axis geomagnetic sensor
Accelerometer	16-bit 3-axis	16-bit 3-axis	16-bit 3-axis	❌
Gyroscope	16-bit 3-axis	16-bit 3-axis	16-bit 3-axis	❌
Magnetometer	❌	BMM350 (TMR, approx. 0.1 μT)	BMM350 (TMR, approx. 0.1 μT)	BMM350 (TMR, approx. 0.1 μT)
Barometric Pressure Sensor	❌	❌	SPA06-003	❌
I2C Address	0x68 / 0x69 (onboard switch)	BMI270: 0x68 / 0x69 (onboard switch) BMM350: 0x14	BMI270: 0x68 / 0x69 (onboard switch) BMM350: 0x14 SPA06-003: 0x76	0x14
Main Functions	Acceleration, angular velocity, and temperature acquisition; on-chip motion detection	Acceleration, angular velocity, temperature, and three-axis magnetic field acquisition; on-chip motion detection	Acceleration, angular velocity, temperature, three-axis magnetic field, and pressure acquisition; on-chip motion detection	Three-axis magnetic field acquisition; heading output after calibration